

Table 1: Performance of DetectGPT and Fast-DetectGPT concerning TPR@FPR-0.5%/100. The detection model is trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	avg
Essay	DetectGPT	37.04 $\pm$ 10.21	24.75 $\pm$ 10.03	5.52 $\pm$ 1.93	21.45 $\pm$ 14.25	15.45 $\pm$ 7.05	4.48 $\pm$ 2.57	18.12
	Fast-DetectGPT	9.92 $\pm$ 11.83	37.38 $\pm$ 23.92	51.52 $\pm$ 24.00	15.66 $\pm$ 24.11	0.09 $\pm$ 0.17	0.96 $\pm$ 1.55	19.25
Reuters	DetectGPT	4.40 $\pm$ 2.62	0.64 $\pm$ 0.54	2.32 $\pm$ 1.87	2.56 $\pm$ 2.80	0.48 $\pm$ 0.47	3.04 $\pm$ 1.61	2.24
	Fast-DetectGPT	72.00 $\pm$ 5.47	9.28 $\pm$ 2.44	94.08 $\pm$ 0.82	93.04 $\pm$ 2.10	1.20 $\pm$ 0.91	9.04 $\pm$ 3.23	46.44
SQuAD1	DetectGPT	0.35 $\pm$ 0.69	0.36 $\pm$ 0.48	0.47 $\pm$ 0.44	0.58 $\pm$ 0.73	0.58 $\pm$ 0.37	0.81 $\pm$ 0.28	0.52
	Fast-DetectGPT	2.66 $\pm$ 2.12	3.45 $\pm$ 1.97	0.94 $\pm$ 0.47	1.73 $\pm$ 1.21	2.67 $\pm$ 1.14	18.27 $\pm$ 7.39	4.95

Table 2: Performance of DetectGPT and Fast-DetectGPT concerning AUROC/100. The detection model is trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	avg
Essay	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	Fast-DetectGPT	96.41 $\pm$ 0.96	94.82 $\pm$ 1.19	99.20 $\pm$ 0.22	97.91 $\pm$ 0.49	84.16 $\pm$ 1.97	89.19 $\pm$ 0.76	93.61
Reuters	DetectGPT	92.78 $\pm$ 1.34	85.71 $\pm$ 2.02	49.24 $\pm$ 1.65	91.59 $\pm$ 1.67	83.41 $\pm$ 1.73	66.59 $\pm$ 2.53	78.22
	Fast-DetectGPT	97.70 $\pm$ 0.16	70.76 $\pm$ 1.59	99.11 $\pm$ 0.12	99.05 $\pm$ 0.20	51.26 $\pm$ 1.29	83.49 $\pm$ 1.70	83.56
SQuAD1	DetectGPT	52.60 $\pm$ 2.74	47.47 $\pm$ 1.42	52.05 $\pm$ 1.87	50.88 $\pm$ 2.47	50.24 $\pm$ 2.20	51.12 $\pm$ 2.44	50.73
	Fast-DetectGPT	65.04 $\pm$ 0.10	63.47 $\pm$ 1.23	62.18 $\pm$ 0.40	61.24 $\pm$ 0.61	63.64 $\pm$ 0.78	77.74 $\pm$ 0.55	65.55

Table 3: Performance comparison (TPR@FPR-0.5%/100) in DetectRL dataset. The detection model is trained on Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	87.37 $\pm$ 8.94	67.58 $\pm$ 11.18	69.64 $\pm$ 13.03	99.12 $\pm$ 0.68	80.93
ChatGPT-STK	90.11$\pm$4.85	71.73$\pm$7.41	73.56$\pm$12.67	99.21$\pm$0.51	83.65
OpenAI-D	99.00$\pm$0.26	68.59 $\pm$ 2.56	98.94 $\pm$ 0.64	99.91$\pm$0.09	91.61
OpenAI-STK	98.98 $\pm$ 0.28	70.96$\pm$3.15	99.08$\pm$0.66	99.91$\pm$0.09	92.23
MPU	96.39 $\pm$ 1.43	86.51 $\pm$ 1.70	99.38 $\pm$ 0.73	99.87 $\pm$ 0.08	95.54
MPU-STK	96.41$\pm$1.33	88.41$\pm$0.85	99.41$\pm$0.59	99.91$\pm$0.08	96.03

Table 4: Performance comparison (AUROC/100) in DetectRL dataset. The detection model is trained on Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	99.06 $\pm$ 0.43	96.72 $\pm$ 1.39	99.31 $\pm$ 0.42	99.94$\pm$0.04	98.76
ChatGPT-STK	99.18$\pm$0.32	96.89$\pm$0.92	99.48$\pm$0.24	99.94$\pm$0.04	98.87
OpenAI-D	99.91$\pm$0.02	95.57 $\pm$ 0.63	99.79 $\pm$ 0.05	100.00$\pm$0.01	98.82
OpenAI-STK	99.91$\pm$0.02	96.20$\pm$0.76	99.81$\pm$0.08	100.00$\pm$0.01	98.98
MPU	99.81$\pm$0.08	99.24 $\pm$ 0.11	99.94$\pm$0.04	100.00$\pm$0.00	99.74
MPU-STK	99.80 $\pm$ 0.07	99.32$\pm$0.11	99.94$\pm$0.04	100.00$\pm$0.00	99.77

Table 5: Performance comparison (TPR@FPR-0.5%/100) against paraphrase attack Polishing. The detection model is trained on Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	avg.
ChatGPT-D	85.73 $\pm$ 7.41	70.11 $\pm$ 10.93	94.90 $\pm$ 3.18	97.66 $\pm$ 1.74	87.10
ChatGPT-STK	87.95 $\pm$ 7.85	76.33 $\pm$ 15.11	92.29 $\pm$ 8.72	97.67 $\pm$ 2.94	88.56
OpenAI-D	99.70 $\pm$ 0.18	89.59 $\pm$ 1.95	99.98 $\pm$ 0.03	100.00 $\pm$ 0.00	97.32
OpenAI-STK	99.73 $\pm$ 0.19	89.92 $\pm$ 1.64	99.98 $\pm$ 0.03	100.00 $\pm$ 0.00	97.41
MPU	98.60 $\pm$ 0.27	95.26 $\pm$ 1.10	99.97 $\pm$ 0.04	100.00 $\pm$ 0.00	98.46
MPU-STK	98.52 $\pm$ 0.21	95.46 $\pm$ 1.13	99.97 $\pm$ 0.04	100.00 $\pm$ 0.00	98.49

Table 6: Performance comparison (TPR@FPR-0.5%/100) against paraphrase attack Back-translation. The detection model is trained on Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	87.88 $\pm$ 6.59	85.35 $\pm$ 6.26	95.90 $\pm$ 3.39	97.21 $\pm$ 2.21	91.58
ChatGPT-STK	90.06 $\pm$ 2.35	87.22 $\pm$ 2.71	96.67 $\pm$ 2.03	97.85 $\pm$ 1.03	92.95
OpenAI-D	99.64 $\pm$ 0.08	99.58 $\pm$ 0.18	99.48 $\pm$ 0.15	99.40 $\pm$ 0.29	99.52
OpenAI-STK	99.62 $\pm$ 0.13	99.59 $\pm$ 0.17	99.50 $\pm$ 0.21	99.40 $\pm$ 0.25	99.53
MPU	95.90 $\pm$ 0.71	94.25 $\pm$ 1.19	99.78 $\pm$ 0.12	99.78 $\pm$ 0.13	97.43
MPU-STK	95.97 $\pm$ 0.79	94.35 $\pm$ 1.25	99.78 $\pm$ 0.12	99.78 $\pm$ 0.13	97.47

Table 7: Cross-domain performance concerning TPR@FPR-0.5%/100. The detection model is trained on ChatGPT text, with the source domain as Essay and the target domain as Reuters.

Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg.
ChatGPT-D	90.64 $\pm$ 3.07	83.84 $\pm$ 2.97	84.40 $\pm$ 9.06	97.28 $\pm$ 0.64	39.76 $\pm$ 5.36	2.48 $\pm$ 1.48	66.40
ChatGPT-STK	94.48 $\pm$ 1.80	86.08 $\pm$ 1.22	87.52 $\pm$ 4.90	98.00 $\pm$ 0.84	45.28 $\pm$ 4.75	10.00 $\pm$ 3.23	70.23
OpenAI-D	79.52 $\pm$ 35.36	68.40 $\pm$ 20.47	78.96 $\pm$ 33.48	83.60 $\pm$ 28.81	37.52 $\pm$ 11.76	9.84 $\pm$ 2.30	59.64
OpenAI-STK	80.80 $\pm$ 23.55	66.40 $\pm$ 17.49	82.08 $\pm$ 21.02	88.80 $\pm$ 13.98	33.76 $\pm$ 11.95	16.16 $\pm$ 3.87	61.33
MPU	100.00 $\pm$ 0.00	87.92 $\pm$ 1.37	89.84 $\pm$ 4.19	99.28 $\pm$ 0.30	55.84 $\pm$ 3.28	85.52 $\pm$ 3.31	86.40
MPU-STK	100.00 $\pm$ 0.00	97.28 $\pm$ 1.74	99.36 $\pm$ 1.28	99.52 $\pm$ 0.59	74.24 $\pm$ 12.40	82.64 $\pm$ 15.34	92.17

Table 8: Cross-domain performance concerning AUROC%/100. The detection model is trained on ChatGPT text, with the source domain as Essay and the target domain as Reuters.

Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg.
ChatGPT-D	98.83 $\pm$ 0.60	97.39 $\pm$ 1.11	97.67 $\pm$ 1.36	99.42 $\pm$ 0.22	80.18 $\pm$ 1.55	42.99 $\pm$ 7.00	86.08
ChatGPT-STK	99.27 $\pm$ 0.59	97.56 $\pm$ 1.41	98.07 $\pm$ 0.69	99.39 $\pm$ 0.27	84.41 $\pm$ 4.53	63.22 $\pm$ 12.01	90.32
OpenAI-D	99.46 $\pm$ 0.67	99.03 $\pm$ 0.68	99.44 $\pm$ 0.57	99.52 $\pm$ 0.56	96.15 $\pm$ 0.81	82.63 $\pm$ 1.73	96.04
OpenAI-STK	99.54 $\pm$ 0.25	98.93 $\pm$ 0.51	99.44 $\pm$ 0.27	99.45 $\pm$ 0.38	95.38 $\pm$ 0.96	86.72 $\pm$ 2.16	96.57
MPU	100.00 $\pm$ 0.00	99.39 $\pm$ 0.09	99.60 $\pm$ 0.21	99.97 $\pm$ 0.02	95.97 $\pm$ 0.92	99.48 $\pm$ 0.12	99.07
MPU-STK	100.00 $\pm$ 0.00	99.78 $\pm$ 0.10	99.91 $\pm$ 0.19	99.99 $\pm$ 0.01	97.94 $\pm$ 1.06	99.34 $\pm$ 0.44	99.49